



NORTH SYDNEY BOYS HIGH SCHOOL

2014

HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK 1

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- All necessary working should be shown
in every question
- Each new question is to be started on a
new page

Class teacher:

(Please tick or highlight)

- ☐ 12M2A Mr Lin
- ☐ 12M2B Ms Juhn
- ☐ 12M3A Mr Zuber
- ☐ 12M3B Mr Berry
- ☐ 12M3C Mr Lowe
- ☐ 12M4A Ms Ziazaris
- ☐ 12M4B Mr Lam
- ☐ 12M4C Mr Ireland

Student Number:

(To be used by examination markers only)

	Section I	Question 7	Question 8	Question 9	Question 10	Total	%
Mark	<u>6</u>	<u>13</u>	<u>11</u>	<u>13</u>	<u>10</u>	<u>53</u>	

Q10 -- give additional 5 min.

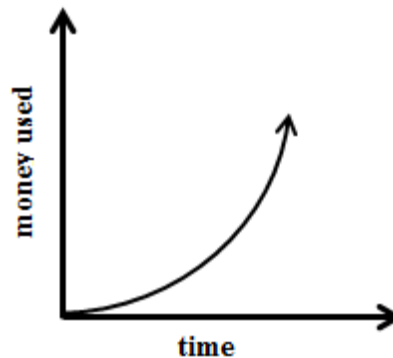
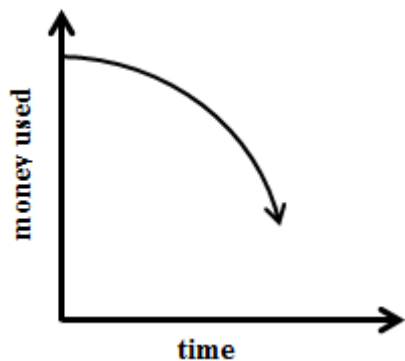
Section I

6 marks

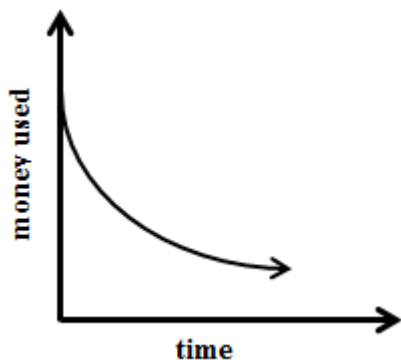
Attempt Questions 1-6

Use the multiple-choice answer sheet for Questions 1-6.

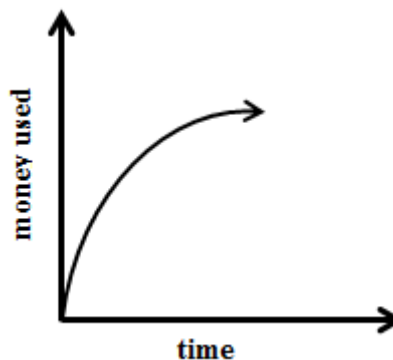
- 1 The sequence $-2, x, 14, \dots$ form an arithmetic sequence. What is the value of x ?
(A) 4 (B) 8
(C) -8 (D) 6
- 2 For the geometric series $5, -1, \frac{1}{5}, -\frac{1}{25}, \dots$, what formula would you use to find the sum of n terms?
(A) $S_n = 2a + (n-1)d$ (B) $S_n = \frac{n}{2}(2a + (n-1)d)$
(C) $S_n = \frac{a(1-r^n)}{1-r}$ (D) $S_n = \frac{a}{1-r}$
- 3 Jamie has been using his money at an increasing rate. Which graph best depicts this information?
(A) (B)



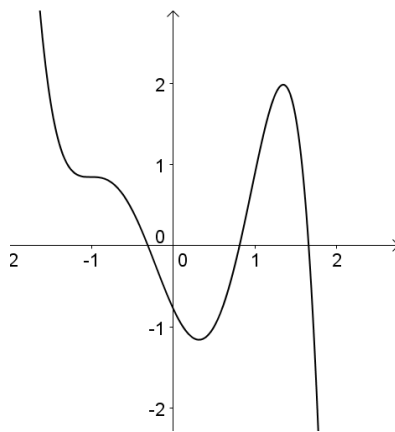
(C)



(D)



- 4 What condition must the discriminant (Δ) satisfy in order for the quadratic equations $ax^2 + bx + c = 0$ have real roots
(A) $\Delta > 0$ (B) $\Delta < 0$
(C) $\Delta = 0$ (D) None of these



Use the diagram above for questions 5 and 6.

- 5 How many stationary points are there (if any) in the graph?
 (A) 2 (B) 3
 (C) 4 (D) There are no stationary points.
- 6 How many points of inflexion are there (if any) in the graph?
 (A) 2 (B) 3
 (C) 4 (D) There are no points of inflexion.

Section II

49 marks

Attempt Questions 7-10

Question 7 (13 marks) Use a NEW page.

- (a) Consider the sequence 98, 92, 86, 80, ...
 (i) Find an expression for the n^{th} term in simplest form. 2
 (ii) Find the sum of the first 50 terms. 2
 (iii) What is the value of the first negative term of the sequence? 3
- (b) Consider the sequence $1, (3x + 2), (3x + 2)^2, \dots$
 (i) For what values of x does the sequence have a limiting sum? 2
 (ii) If $x = -\frac{1}{2}$, find the limiting sum. 2
- (c) Evaluate $\sum_{k=6}^8 (2k + 1)(1 - k)$ 2

Question 8 (11 marks) Use a NEW page.

- (a) For what values of x is the function $f(x) = -4x^3 + 27x^2 + 30x - 18$ decreasing? 3
- (b) A function is given by $f(x) = 3x^4 - 16x^3 + 24x^2$.
 (i) Find any stationary points and determine their nature. 2
 (ii) Find any points of inflexion. 2
 (iii) Sketch the curve, clearly showing the intercepts with axes, turning points and points of inflexion. 3
 (iv) What is the minimum value of $f(x)$? 1

Question 9 continues over the page

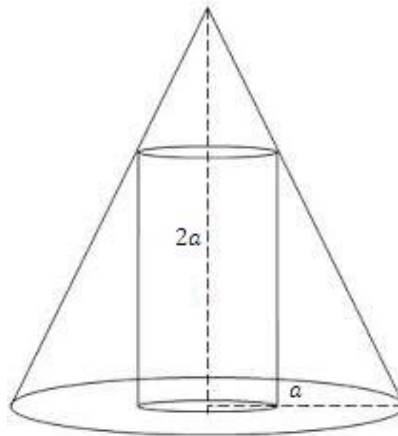
Question 9 (13 marks) Use a NEW page.

- (a) For the equation $12(2^{2x}) + (2^x) - 1 = 0$, solve for x . 3
- (b) α and β are the roots to the quadratic equation $2x^2 - 4x + 6 = 0$. Find:
- (i) $\alpha + \beta$ 1
 - (ii) $\alpha\beta$ 1
 - (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2
 - (iv) $\alpha^2 + \beta^2$ 2
- (c) For what value(s) of b will the equation $x^2 + 4x - 2b = 0$ have:
- (i) one of the roots equal to 3. 1
 - (ii) roots which are reciprocals of each other. 1
 - (iii) the sum of the roots equal to their product. 2

Question 10 (10 marks) Use a NEW page.

- (a) Sketch the curve satisfying the following conditions, clearly showing the important features. 3
- $f(x) > 0$ for all values of x and $f(0) = 1$
 - $f'(x) = 0$ at $x = 2$, $f'(x) > 0$ for every other values of x
 - $f''(x) > 0$ for $x < 1$ or $x > 2$, $f''(x) < 0$ for $1 < x < 2$
 - $f''(x) = 0$ at $x = 1$ and $x = 2$

(b)



A cylinder is inscribed in a cone of base radius a and height $2a$. The radius of the cylinder is given by r and h is the height of the cone.

- (i) Show that $r = a - \frac{h}{2}$. 1
 - (ii) Show that the volume of the cylinder is $V = \pi(a^2h - ah^2 + \frac{h^3}{4})$ 1
 - (iii) Find the maximum volume of the cylinder in term of a in simplest form. 3
- (c) Find the sum of the following infinite series. 2

$$\frac{1}{6} + \frac{13}{6^2} + \frac{19}{6^3} + \frac{97}{6^4} + \dots + \frac{3^n + (-2)^n}{6^n} + \dots$$

End of paper

SUGGESTED SOLUTIONS.

Multiple choice

1. D
2. C
3. B
4. D
5. B
6. B

7. 98, 92, 86, 80, ...

(a) (i) $a = 98$ $d = -6$

$$\begin{aligned}T_n &= 98 + (n-1)(-6) \\&= 98 - 6n + 6 \\&= 104 - 6n\end{aligned}$$

(ii) $S_{50} = 25(2 \times 98 + (49)(-6))$
 $= -2450.$

(iii) $104 - 6n < 0$

$$6n > 104$$

$$n > 17.\bar{3}$$

18th term.

(b) (i) $-1 < 3x + 2 < 1$

$$-3 < 3x < -1$$

$$-1 < x < -\frac{1}{3}$$

(ii) when $x = -\frac{1}{2}$, $S_{\infty} = \frac{1}{1 - (-\frac{3}{2} + 2)}$
 $= 2$

(c) $(-65) + (-90) + (-119)$

$$= -274$$

8. (a) $f(x) = -4x^3 + 27x^2 + 30x - 18$

$$f'(x) = -12x^2 + 54x + 30$$

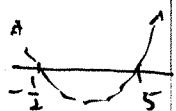
decreasing when $f'(x) < 0$

$$-12x^2 + 54x + 30 < 0$$

$$2x^2 - 9x - 5 > 0$$

$$(2x + 1)(x - 5) > 0$$

$$x < -\frac{1}{2} \text{ or } x > 5$$



(b) $f(x) = 3x^4 - 16x^3 + 24x^2$

(i) $f'(x) = 12x^3 - 48x^2 + 48x$

$$f''(x) = 36x^2 - 96x + 48$$

Stationary points occur when $f'(x) = 0$

$$x(x^2 - 4x + 4) = 0$$

$$x(x-2)^2 = 0$$

$$\therefore x = 0 \quad x = 2$$

To determine the nature of stationary points

$$f''(0) = 48 \text{ which is } > 0$$

$\therefore$ minimum turning point at $(0, 0)$

$$f''(2) = 0$$

$\therefore$ possible point of inflexion at $(2, 16)$

(ii) Points of inflexion occur when $f''(x) = 0$

$$36x^2 - 96x + 48 = 0$$

$$3x^2 - 8x + 4 = 0$$

$$(3x - 2)(x - 2) = 0$$

$$x = \frac{2}{3} \quad x = 2$$

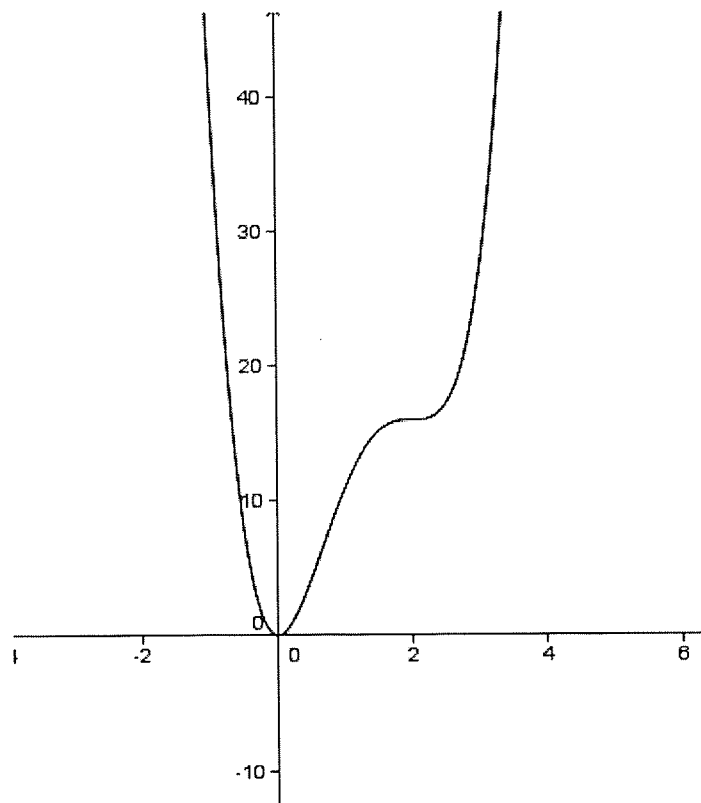
To see if there is a change of concavity

x	$\frac{2}{3}$	2
$f''(x)$		

(Various solutions)

$\therefore$ points of inflexion at $(\frac{2}{3}, \frac{176}{27})$ and $(2, 16)$

(iii)



civ) minimum value is 0

$$9. (a) 12(2^{2x}) + (2^x) - 1 = 0$$

$$\text{let } u = 2^x$$

$$(2u^2 + u - 1 = 0$$

$$(3u + 1)(4u - 1) = 0$$

$$u = -\frac{1}{3} \quad u = \frac{1}{4}$$

$$2^x = -\frac{1}{3}$$

no real solutions

$$2^x = \frac{1}{4}$$

$$\therefore x = -2$$

$$(b) 2x^2 - 4x + 6 = 0$$

$$(i) \alpha + \beta = 2$$

$$(ii) \alpha\beta = 3$$

$$(iii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{2}{3}$$

$$\begin{aligned} (iv) \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= 4 - 6 \\ &= -2 \end{aligned}$$

$$(c) x^2 + 4x - 2b = 0$$

$$(i) \text{ let } x = 3 \quad 9 + 12 - 2b = 0$$

$$b = \frac{21}{2}$$

$$(ii) \alpha, \frac{1}{\alpha}$$

$$\alpha \times \frac{1}{\alpha} = -2b$$

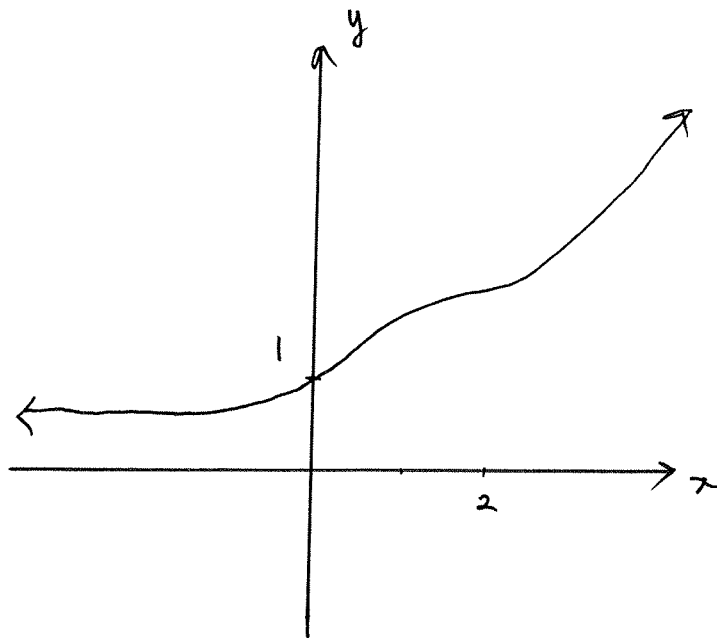
$$\therefore b = \frac{1}{-2}$$

$$(iii) \alpha + \beta = \alpha\beta$$

$$-4 = -2b$$

$$\therefore b = 2$$

10, (b)



$$(i) \quad (i) \quad \frac{r}{2a-h} = \frac{a}{2a}$$

$$\therefore r = a - \frac{h}{2}$$

$$\begin{aligned} (ii) \quad V &= \pi \left(a - \frac{h}{2}\right)^2 h \\ &= \pi h \left(a^2 - ah + \frac{h^2}{4}\right) \\ &= \pi \left(a^2 h - ah^2 + \frac{h^3}{4}\right) \end{aligned}$$

$$(iii) \quad \frac{dV}{dh} = a^2 \pi - 2a\pi h + \frac{3\pi h^2}{4}$$

$$\text{let } \frac{dV}{dh} = 0$$

$$a^2 - 2ah + \frac{3h^2}{4} = 0$$

$$4a^2 - 8ah + 3h^2 = 0$$

$$(2a - 3h)(2a - h) = 0$$

$$h = \frac{2a}{3} \quad h = 2a$$

$$\frac{d^2V}{dh^2} = -2a\pi + \frac{3\pi h}{2}$$

$$\text{When } h = \frac{2a}{3}, \quad \frac{d^2V}{dh^2} = -2a\pi + \pi a$$

$$= -a\pi \text{ which is } < 0 \quad \therefore \text{maximum at } h = \frac{2a}{3}$$

$$\text{When } h = 2a, \quad \frac{d^2V}{dh^2} = -2a\pi + 3\pi a$$

$$= \pi a \text{ which is } > 0 \quad \therefore \text{minimum at } h = 2a$$

$\therefore$ maximum volume occurs when $h = \frac{2a}{3}$

$$V = \frac{8\pi a^3}{27}$$

$$\begin{aligned}
 (c) \quad & \sum_{n=1}^{\infty} \left(\frac{3}{6}\right)^n + \left(\frac{-2}{6}\right)^n \\
 &= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n + \left(-\frac{1}{3}\right)^n \\
 &= \frac{\frac{1}{2}}{1 - \frac{1}{2}} + \frac{-\frac{1}{3}}{1 + \frac{1}{3}} \\
 &= 1 - \frac{1}{4} \\
 &= \frac{3}{4}
 \end{aligned}$$